localCloudManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"localCloudManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `localCloudManagement` service of the LocalCloudManager system.

Contents

1 Overview **4**
1.1 Interface registration . . . 4

2 Service Operations **6**
2.1 operation session . . . 6
2.2 operation login . . . 6
2.3 operation logout . . . 6
2.4 operation overview . . . 7
2.5 operation serviceRegistryManagement . . . 7
2.6 operation dataManager . . . 7
2.7 operation consumers . . . 8
2.8 operation interCloud . . . 8
2.9 operation sysMonitorHistory . . . 8
2.10 operation sysMonitorSubscriptions . . . 9
2.11 operation subscribeSysMonitor . . . 9
2.12 operation unsubscribeSysMonitor . . . 10
2.13 operation plantDescriptions . . . 10
2.14 operation createPlantDescription . . . 10
2.15 operation replacePlantDescription . . . 11
2.16 operation updatePlantDescription . . . 11
2.17 operation removePlantDescription . . . 11
2.18 operation orchestrationRules . . . 12
2.19 operation createOrchestrationRule . . . 12
2.20 operation removeOrchestrationRule . . . 13
2.21 operation logs . . . 13
2.22 operation lifecycle . . . 13
2.23 operation features . . . 14
2.24 operation authorizationPolicies . . . 14
2.25 operation grantAuthorizationPolicy . . . 15
2.26 operation revokeAuthorizationPolicy . . . 15
2.27 operation changeOrchestrationRulePriority . . . 15
2.28 operation sosConfigurations . . . 16
2.29 operation createSoS . . . 16
2.30 operation replaceSoS . . . 16
2.31 operation removeSoS . . . 17
2.32 operation localCloudConfiguration . . . 17
2.33 operation chipletFloorplans . . . 17

3 Data Models **19**
3.1 struct LifecycleResult . . . 19
3.2 struct LogLines . . . 19
3.3 struct Login . . . 19
3.4 struct OrchestrationRule . . . 19
3.5 struct PlantDescription . . . 19
3.6 struct Session . . . 19
3.7 struct SysMonitorSubscriptionRequest . . . 20
3.8 struct View . . . 20
3.9 struct Features . . . 20
3.10 struct ManagementPolicy . . . 20
3.11 struct PolicyChange . . . 20
3.12 struct PolicyView . . . 21
3.13 struct Removal . . . 21
3.14 struct RulePriorityChange . . . 21

3.15 struct SoSChange 21
3.16 Primitives 21

4 References **23**

5 Revision History **24**
5.1 Amendments 24
5.2 Quality Assurance 24

1 Overview

This document describes the `localCloudManagement` service interface of the LocalCloudManager system, which provides the operator console API of the local cloud: overview, registry and data views, orchestration rules, consumers, inter-cloud state, sysMonitor history and subscriptions, plant descriptions, logs and container lifecycle.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `localCloudManagement` service interface

This document provides the Interface Design Description IDD to the *localCloudManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `localCloudManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_http` and policy `NONE`: the console API is reached through the nginx UI on port 8490 (loopback of the host); the API itself listens on 8491. Base path `/api`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`session`	GET	`/session`
`login`	POST	`/session`
`logout`	DELETE	`/session`
`overview`	GET	`/overview`
`serviceRegistry Management`	GET	`/service-registry-management`
`dataManager`	GET	`/datamanager`
`consumers`	GET	`/consumers`
`interCloud`	GET	`/intercloud`
`sysMonitorHistory`	GET	`/sysmonitor`
`sysMonitorSubscriptions`	GET	`/sysmonitor/subscriptions`
`subscribeSysMonitor`	PUT	`/sysmonitor/subscriptions`
`unsubscribeSysMonitor`	DELETE	`/sysmonitor/subscriptions`
`plantDescriptions`	GET	`/plantdescriptions`
`createPlantDescription`	POST	`/plantdescriptions`

Operation	Method	Path
`replacePlantDescription`	PUT	`/plantdescriptions`
`updatePlantDescription`	PATCH	`/plantdescriptions`
`removePlantDescription`	DELETE	`/plantdescriptions`
`orchestrationRules`	GET	`/orchestration/rules`
`createOrchestrationRule`	POST	`/orchestration/rules`
`removeOrchestrationRule`	DELETE	`/orchestration/rules`
`logs`	GET	`/logs`
`lifecycle`	POST	`/containers`

2 Service Operations

The interfaces of the `localCloudManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `localCloudManagement` SD document, as well as input and output types.

Requests need the console session cookie (from `login`), an allowed `Host`/`Origin`, and for `POST`, `PUT`, `PATCH` and `DELETE` the header `X-Arrowhead-Action: confirm`. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with `{"error": "<text>"}` and the status codes `400 Bad Request` (invalid input), `401 Unauthorized` (no valid identity or session), `403 Forbidden` (caller not permitted, request not forwarded by the console's nginx, or host/origin not allowed), `405 Method Not Allowed`, `409 Conflict` (stored entry changed since it was read), `415 Unsupported Media Type` (login without JSON), `404 Not Found`, `428 Precondition Required` (no confirmation header), `429 Too Many Requests` (login locked), `500 Internal Server Error`, `502 Bad Gateway` (a system it depends on failed) and `503 Service Unavailable` (overloaded, console password not configured, or a provider not registered).

2.1 GET /api/session

Operation: session
Output: Session

Returns whether the console session is valid (no session needed).

A successful call returns `200 OK` with the output in the body.

```
GET /api/session HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 1: An session invocation.

```
{
  "authenticated": true,
  "configured": true
}
```

Listing 2: An session response.

2.2 POST /api/session

Operation: login
Input: Login

Logs in with the console password; sets the HttpOnly, SameSite=Strict session cookie.

A successful call returns `204 No Content`.

```
POST /api/session HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "password": "<console password>"
}
```

Listing 3: An login invocation.

2.3 DELETE /api/session

Operation: logout

Ends the console session and clears the cookie.
A successful call returns `200 OK` with `{"authenticated": false}`.

```
DELETE /api/session HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 4: An logout invocation.

2.4 GET /api/overview

Operation: overview
Output: View

Overview: containers, registered systems, brokers, sysMonitor flows, activity.
A successful call returns `200 OK` with the output in the body.

```
GET /api/overview HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 5: An overview invocation.

```
{}
```

Listing 6: An overview response.

2.5 GET /api/service-registry-management

Operation: serviceRegistryManagement
Output: View

Registry inventory: devices, systems, services and interface templates.
A successful call returns `200 OK` with the output in the body.

```
GET /api/service-registry-management HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 7: An serviceRegistryManagement invocation.

```
{}
```

Listing 8: An serviceRegistryManagement response.

2.6 GET /api/datamanager?limit=*{limit}*

Operation: dataManager
Output: View

DataManager historian streams and recent records.

Query parameters: *limit* (Integer, optional). List values are repeated or comma-separated.
A successful call returns `200 OK` with the output in the body.

```
GET /api/datamanager?limit=<limit> HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 9: An dataManager invocation.

```
{}
```

Listing 10: An dataManager response.

2.7 GET /api/consumers

Operation: consumers
Output: View

Per provider service, the authorized and the orchestrated consumers.
A successful call returns `200 OK` with the output in the body.

```
GET /api/consumers HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 11: An consumers invocation.

```
{}
```

Listing 12: An consumers response.

2.8 GET /api/intercloud

Operation: interCloud
Output: View

Relay connection, Gatekeeper and GateTunnel sessions.
A successful call returns `200 OK` with the output in the body.

```
GET /api/intercloud HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 13: An interCloud invocation.

```
{}
```

Listing 14: An interCloud response.

2.9 GET /api/sysmonitor?system=*{system}*&limit=*{limit}*

Operation: sysMonitorHistory
Output: View

Monitored systems and their history.

Query parameters: *system* (String, optional); *limit* (Integer, optional). List values are repeated or comma-separated.

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor?system=<system>&limit=<limit> HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 15: An sysMonitorHistory invocation.

```
{}
```

Listing 16: An sysMonitorHistory response.

2.10 GET /api/sysmonitor/subscriptions

Operation: **sysMonitorSubscriptions**
Output: **View**

SysMonitor providers, subscriptions and intake state.

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor/subscriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 17: An sysMonitorSubscriptions invocation.

```
{}
```

Listing 18: An sysMonitorSubscriptions response.

2.11 PUT /api/sysmonitor/subscriptions

Operation: **subscribeSysMonitor**
Input: **SysMonitorSubscriptionRequest**
Output: **View**

Sets the SysMonitor subscription of a provider (appended to the path).

Path parameter: *provider*.

A successful call returns `200 OK` with the output in the body.

```
PUT /api/sysmonitor/subscriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "observationTypes": [
    "*"
  ],
  "enabled": true
}
```

Listing 19: An subscribeSysMonitor invocation.

```
{}
```

Listing 20: An subscribeSysMonitor response.

2.12 DELETE /api/sysmonitor/subscriptions

Operation: unsubscribeSysMonitor
Input: String

Removes the SysMonitor subscription of a provider.
Path parameter: *provider*.
A successful call returns `200 OK` with `{"removed": "<provider>"}`.

```
DELETE /api/sysmonitor/subscriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 21: An unsubscribeSysMonitor invocation.

2.13 GET /api/plantdescriptions

Operation: plantDescriptions
Output: View

Plant descriptions with the registered systems and services offered as choices.
A successful call returns `200 OK` with the output in the body.

```
GET /api/plantdescriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 22: An plantDescriptions invocation.

```
{}
```

Listing 23: An plantDescriptions response.

2.14 POST /api/plantdescriptions

Operation: createPlantDescription
Input: PlantDescription
Output: PlantDescription

Creates a plant description in the PlantDescriptionEngine.
A successful call returns `201 Created` with the output in the body.

```
POST /api/plantdescriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{}
```

Listing 24: An createPlantDescription invocation.

```
{}
```

Listing 25: An createPlantDescription response.

2.15 PUT /api/plantdescriptions

Operation: replacePlantDescription
Input: PlantDescription
Output: PlantDescription

Replaces a plant description (identifier appended).
Path parameter: *id*.
A successful call returns `200 OK` with the output in the body.

```
PUT /api/plantdescriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{}
```

Listing 26: An replacePlantDescription invocation.

```
{}
```

Listing 27: An replacePlantDescription response.

2.16 PATCH /api/plantdescriptions

Operation: updatePlantDescription
Input: PlantDescription
Output: PlantDescription

Updates a plant description (identifier appended).
Path parameter: *id*.
A successful call returns `200 OK` with the output in the body.

```
PATCH /api/plantdescriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{}
```

Listing 28: An updatePlantDescription invocation.

```
{}
```

Listing 29: An updatePlantDescription response.

2.17 DELETE /api/plantdescriptions

Operation: removePlantDescription
Input: String

Removes a plant description (identifier appended).
Path parameter: *id*.
A successful call returns `200 OK` with `{"removed": <id>}`.

```
DELETE /api/plantdescriptions HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 30: An removePlantDescription invocation.

2.18 GET /api/orchestration/rules

Operation: orchestrationRules
Output: View

SimpleStore rules and the dynamic orchestrations of DynamicServiceOrchestration.
A successful call returns `200 OK` with the output in the body.

```
GET /api/orchestration/rules HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 31: An orchestrationRules invocation.

```
{}
```

Listing 32: An orchestrationRules response.

2.19 POST /api/orchestration/rules

Operation: createOrchestrationRule
Input: OrchestrationRule
Output: View

Creates a SimpleStore rule.
A successful call returns `201 Created` with the output in the body.

```
POST /api/orchestration/rules HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "consumer": "TemperatureDisplay",
  "serviceInstanceId": "TemperatureSensor|temperatureReading|1.0.0",
  "priority": 1
}
```

Listing 33: An createOrchestrationRule invocation.

```
{}
```

Listing 34: An createOrchestrationRule response.

2.20 DELETE /api/orchestration/rules

Operation: removeOrchestrationRule
Input: String

Removes a SimpleStore rule (identifier appended, a UUID).
Path parameter: *id*.
A successful call returns `200 OK` with `{"id": "<id>", "removed": true, "at": "<time>"}`.

```
DELETE /api/orchestration/rules HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 35: An removeOrchestrationRule invocation.

2.21 GET /api/logs?service={service}&tail={tail}

Operation: logs
Output: LogLines

Log lines of a container of the cloud.
Query parameters: *service* (String, optional); *tail* (Integer, optional). List values are repeated or comma-separated.
A successful call returns `200 OK` with the output in the body.

```
GET /api/logs?service=<service>&tail=<tail> HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 36: An logs invocation.

```
{
  "service": "datamanager",
  "lines": [
    "..."
  ],
  "collectedAt": "2026-10-22T12:00:00Z"
}
```

Listing 37: An logs response.

2.22 POST /api/containers

Operation: lifecycle
Input: String
Output: LifecycleResult

Starts, stops or restarts a managed container: `POST /api/containers/<service>/<action>`.
Path parameter: *service*, *action*.
A successful call returns `202 Accepted` with the output in the body.

```
POST /api/containers HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 38: An lifecycle invocation.

```
{
  "service": "datamanager",
  "action": "restart",
  "accepted": true,
  "at": "string"
}
```

Listing 39: An lifecycle response.

Console operations outside the registered operation list

The following operations of the same API are not listed in the registered `operations` property; the same session, host, proxy and confirmation rules apply.

2.23 GET /api/features

Operation: features
Output: Features

Which optional console views are enabled.
A successful call returns `200 OK`.

```
GET /api/features HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 40: An features invocation.

```
{
  "localCloudConfiguration": false,
  "chipletFloorplans": false,
  "managerName": "LocalCloudManager"
}
```

Listing 41: An features response.

2.24 GET /api/orchestration/policies

Operation: authorizationPolicies
Output: PolicyView

Local-cloud management policies of ConsumerAuthorization (editable) and provider-owned local policies (read-only).
A successful call returns `200 OK`.

```
GET /api/orchestration/policies HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 42: An authorizationPolicies invocation.

2.25 POST /api/orchestration/policies

Operation: grantAuthorizationPolicy
Input: PolicyChange
Output: ManagementPolicy

Grants a local management policy, or replaces the policy of the same provider and target when `expected` matches the stored one. Provider and service must be registered in the local cloud; policy types are `ALL`, `WHITELIST` and `BLACKLIST`.

A successful call returns `201 Created (200 OK on replacement)`.

```
POST /api/orchestration/policies HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "policy": {
    "provider": "TemperatureSensor",
    "targetType": "SERVICE_DEF",
    "target": "temperatureReading",
    "defaultPolicy": { "policyType": "WHITELIST", "policyList": ["TemperatureDisplay"] },
    "scopedPolicies": {}
  }
}
```

Listing 43: An grantAuthorizationPolicy invocation.

2.26 DELETE /api/orchestration/policies

Operation: revokeAuthorizationPolicy
Input: PolicyChange
Output: Removal

Revokes a local management policy (instance identifier appended to the path) when `expected` matches the stored policy.

A successful call returns `200 OK`.

```
DELETE /api/orchestration/policies HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 44: An revokeAuthorizationPolicy invocation.

```
{
  "removed": true,
  "id": "<instanceId>"
}
```

Listing 45: An revokeAuthorizationPolicy response.

2.27 PUT /api/orchestration/rules

Operation: changeOrchestrationRulePriority
Input: RulePriorityChange
Output: View

Changes the priority of a SimpleStore rule (identifier appended) with SimpleStore `modify-priorities`, or by create and remove when `ORCHESTRATION_PRIORITY_MODE=replace`.

A successful call returns `200 OK`.

```
PUT /api/orchestration/rules HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "priority": 2,
  "expected": { "consumer": "TemperatureDisplay", "serviceInstanceId": "TemperatureSensor|
    temperatureReading|1.0.0", "priority": 1 }
}
```

Listing 46: An changeOrchestrationRulePriority invocation.

2.28 GET /api/sos

Operation: sosConfigurations
Output: View

SoS compositions (plant descriptions of the PlantDescriptionEngine); one composition when its identifier is appended.

A successful call returns `200 OK`.

```
GET /api/sos HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 47: An sosConfigurations invocation.

2.29 POST /api/sos

Operation: createSoS
Input: SoSChange
Output: PlantDescription

Stores a validated SoS composition (unique systems and ports, required-to-provided connections, included compositions without cycles).

A successful call returns `201 Created`.

```
POST /api/sos HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 48: An createSoS invocation.

2.30 PUT /api/sos

Operation: replaceSoS
Input: SoSChange
Output: PlantDescription

Replaces an SoS composition (identifier appended) when `expected` matches the stored one.
A successful call returns `200 OK`.

```
PUT /api/sos HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 49: An replaceSoS invocation.

2.31 DELETE /api/sos

Operation: removeSoS
Input: SoSChange
Output: Removal

Removes an SoS composition (identifier appended) when `expected` matches the stored one.
A successful call returns `200 OK`.

```
DELETE /api/sos HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
X-Arrowhead-Action: confirm
```

Listing 50: An removeSoS invocation.

```
{
  "removed": 3
}
```

Listing 51: An removeSoS response.

2.32 GET /api/localcloud

Operation: localCloudConfiguration
Input: String
Output: View

Pass-through to the LocalCloudConfigurator when `LOCAL_CLOUD_CONFIGURATOR_URL` is set (404 otherwise): `GET` of `/catalog`, `/status`, `/configurations`, `/configurations/{uuid}` and `/deployments/{id}`; `POST` of `/plan`, `/deployments` and `/configurations`; `PUT` and `DELETE` of `/configurations/{uuid}`. Status and body of the configurator are returned unchanged.
A successful call returns `the configurator's status`.

```
GET /api/localcloud HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 52: An localCloudConfiguration invocation.

2.33 GET /api/chiplet/floorplans

Operation: chipletFloorplans
Input: String
Output: View

Chiplet floorplan view when `CHIPLET_FLOORPLANS_ENABLED=true` (404 otherwise), through orchestrated ChipletFloorplanGenerator services: `GET` list (`page`, `size` up to 200), `/library` and `/{id}`; `POST` create (stored inactive, 201) and `POST /check-network`; `PUT /{id}` with `expectedUpdatedAt` (409 when changed); `DELETE /{id}`. Bodies carry `floorplan` with 1–64 chiplets and at most 1024 links.

A successful call returns `200 OK`.

```
GET /api/chiplet/floorplans HTTP/1.1
Host: cloud-monitor-api:8490
Cookie: <console session>
```

Listing 53: An chipletFloorplans invocation.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.16, which are used to represent things like hashes, identifiers and texts.

3.1 struct LifecycleResult

Object Field	Value Type	Description
"service"	String	Compose service.
"action"	String	`start`, `stop` or `restart`.
"accepted"	Boolean	Forwarded to the Docker proxy.
"at"	String	Time of the action.

3.2 struct LogLines

Object Field	Value Type	Description
"service"	String	Compose service.
"lines"	Array<String>	Log lines (`tail` latest).
"collectedAt"	DateTime	Collection time.

3.3 struct Login

Object Field	Value Type	Description
"password"	String	Console password (JSON only; 204 on success, 401 wrong, 429 locked).

3.4 struct OrchestrationRule

Object Field	Value Type	Description
"consumer"	String	Consumer system.
"serviceInstanceId"	String	Service instance.
"priority"	Number	Priority (1 highest).

3.5 struct PlantDescription

Object Field	Value Type	Description
"*"	Object	Plant description as in PlantDescriptionEngine `plantDescriptionManagement`.

3.6 struct Session

Object Field	Value Type	Description
"authenticated"	Boolean	Whether the request carries a valid console session.
"configured"	Boolean	Whether a console password is configured.

3.7 struct SysMonitorSubscriptionRequest

Object Field	Value Type	Description
"observationTypes"	Array<String>	Types to receive, * for all.
"enabled"	Boolean	Default true.

3.8 struct View

Object Field	Value Type	Description
"*"	Object	View document assembled from the providing systems; a failing source is reported in the document instead of failing the request.

3.9 struct Features

Object Field	Value Type	Description
"localCloudConfiguration"	Boolean	Local-cloud configuration view enabled.
"chipletFloorplans"	Boolean	Chiplet floorplan view enabled.
"managerName"	String	LocalCloudManager, or ChipletLocalCloudManager with floorplans.

3.10 struct ManagementPolicy

Object Field	Value Type	Description
"instanceId"	String	Policy instance (assigned by Consumer Authorization).
"provider"	String	Registered local provider.
"targetType"	String	`SERVICE_DEF` or `EVENT_TYPE`.
"target"	String	Service definition or event type.
"description"	String	At most 1000 characters.
"defaultPolicy"	Object	`policyType` (`ALL`, `WHITELIST`, `BLACKLIST`) and `policyList` of local system names.
"scopedPolicies"	Object	Per operation, a policy like `defaultPolicy` (at most 100).

3.11 struct PolicyChange

Object Field	Value Type	Description
"policy"	ManagementPolicy	Policy to grant (POST).
"expected"	ManagementPolicy	Stored policy as last read; required to replace or revoke.

3.12 struct PolicyView

Object Field	Value Type	Description
"entries"	Array	Editable local management policies.
"readOnlyEntries"	Array	Provider-owned local policies.
"collectedAt"	DateTime	Collection time.

3.13 struct Removal

Object Field	Value Type	Description
"removed"	Object	Identifier of, or `true` for, the removed entry.
"id"	String	Removed policy (policies only).

3.14 struct RulePriorityChange

Object Field	Value Type	Description
"priority"	Number	New priority, 1 to 2147483647.
"expected"	OrchestrationRule	Rule as last read (409 if changed, 428 if missing).

3.15 struct SoSChange

Object Field	Value Type	Description
"configuration"	Object	SoS model: `plantDescription` (name), `active`, `include`, `systems` with ports, `connections`.
"expected"	Object	Stored composition as last read (replace and remove).
"activationConfirmed"	Boolean	Must be `true` to save an active composition (428 otherwise).

3.16 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “localCloudManagement – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console operations outside the registered list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) added; success responses of logout, unsubscribeSysMonitor, removePlantDescription and removeOrchestrationRule corrected; error codes completed.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	